

TINDAL, Australia

WMO: 941310

Lat: 14.5228S

Lon: 132.3825E

Elev: 135

StdP: 99.71

Time Zone: 9.50 (AUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	9.8	11.2	-6.8	2.2	26.5	-4.0	2.7	24.5	8.3	25.0	7.6	25.5	2.4	130	0.389

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	11.9	39.0	21.5	38.0	21.8	37.1	22.1	28.8	31.5	26.9	31.8	26.2	31.2	3.9	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
28.2	24.8	29.7	25.9	21.6	28.6	25.1	20.6	27.8	95.0	32.5	85.7	31.9	82.9	31.6	37.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.7	6.9	6.2	6.4	40.3	1.9	1.0	5.0	41.0	3.9	41.6	2.9	42.2	1.5	42.9		
(5)			3.8	31.2	1.4	3.5	2.8	33.8	2.0	35.8	1.2	37.8	0.2	40.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.0	29.0	28.9	28.5	27.1	24.6	22.0	21.9	23.2	27.6	30.4	30.9	30.1		
	DBStd	3.80	1.62	1.44	1.39	1.82	2.61	2.90	2.46	2.54	2.50	1.90	1.81	1.83		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	9	0	0	0	0	1	5	3	1	0	0	0	0		
	CDD10.0	6202	587	528	572	514	452	360	368	408	527	634	627	623		
	CDD18.3	3169	329	295	314	264	193	115	113	151	277	376	377	365		
	CDH23.3	39056	3646	3240	3464	3033	2149	1289	1406	2028	3786	5377	5143	4496		
	CDH26.7	20128	1676	1496	1656	1545	994	497	556	961	2143	3220	3014	2371		
(14) Wind	WSAvg	2.8	2.7	2.6	2.4	2.8	3.1	3.2	3.2	3.0	2.8	2.9	2.7	2.6		
(15) Precipitation	PrecAvg	1100	267	210	179	37	3	1	0	2	6	30	91	270		
	PrecMax	1591	677	512	417	192	26	7	2	30	57	111	260	567		
	PrecMin	626	97	36	11	0	0	0	0	0	0	0	11	103		
	PrecStd	258	140	132	121	54	6	2	0	6	13	38	58	131		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.1	37.1	37.1	36.2	34.9	33.3	33.5	35.2	37.9	40.0	40.2	40.1		
		MCWB	25.1	23.7	23.7	22.4	21.7	20.5	19.8	19.4	20.1	21.0	21.1	22.6		
	2%	DB	35.9	36.0	35.8	35.1	33.8	32.1	32.1	34.0	37.0	38.9	39.1	38.0		
		MCWB	25.1	24.4	23.7	22.2	21.3	19.9	18.8	18.8	19.8	21.0	21.3	24.0		
	5%	DB	34.9	34.9	34.7	34.2	32.8	31.0	31.1	32.9	36.1	37.9	38.0	36.8		
		MCWB	25.4	24.8	23.7	22.0	21.0	19.4	18.1	18.1	19.5	21.0	21.8	24.6		
	10%	DB	33.8	33.8	33.8	33.3	31.8	29.9	30.0	31.6	35.1	36.9	36.9	35.1		
		MCWB	25.5	25.1	23.7	21.9	20.5	18.5	17.5	17.4	19.2	20.9	22.6	24.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.4	28.3	31.0	28.3	28.1	23.2	22.7	22.4	26.2	33.1	32.0	32.5		
		MCDB	33.7	31.6	31.5	29.3	29.6	28.4	29.3	29.6	32.4	34.9	32.0	33.3		
	2%	WB	27.0	26.8	26.8	25.2	24.1	21.6	21.0	21.1	23.6	25.8	26.0	26.7		
		MCDB	32.3	31.6	31.3	29.5	29.6	29.3	28.8	29.7	29.8	30.8	31.9	32.1		
	5%	WB	26.4	26.3	26.0	24.6	23.1	20.7	19.9	20.2	22.7	24.2	25.2	26.1		
		MCDB	31.5	31.0	30.4	29.1	29.2	28.4	28.0	29.4	30.2	30.7	31.4	31.6		
	10%	WB	26.0	25.9	25.5	24.1	22.2	19.8	18.9	19.2	22.0	23.7	24.7	25.7		
		MCDB	30.8	30.4	29.7	28.7	28.5	27.4	27.3	28.7	30.3	30.4	30.9	31.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.1	9.3	10.3	12.7	14.1	15.0	16.2	16.9	15.4	13.2	11.9	10.4		
		MCDBR	10.9	11.3	12.4	13.9	14.1	14.9	16.5	17.3	15.4	14.3	13.2	12.2		
	5% WB	MCWBR	3.4	3.4	3.6	4.0	4.7	5.2	6.1	6.0	5.3	5.5	4.5	3.5		
		MCWBR	9.4	9.2	9.4	10.4	12.1	13.7	14.7	14.5	12.7	12.0	11.2	10.1		
(40) Clear-Sky Solar Irradiance	taub	taub	0.407	0.403	0.390	0.369	0.356	0.332	0.327	0.342	0.406	0.435	0.450	0.425		
		taud	2.496	2.503	2.535	2.562	2.537	2.552	2.559	2.488	2.325	2.277	2.292	2.421		
	Ebn at Noon	Ebn at Noon	938	934	926	912	891	898	912	925	896	893	892	919		
		Edh at Noon	116	114	108	99	95	91	92	105	131	142	141	125		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	6.03	6.12	6.03	5.94	5.40	5.15	5.45	6.18	6.76	7.00	6.75	6.39		
	RadStd	RadStd	0.54	0.72	0.64	0.45	0.31	0.25	0.16	0.19	0.34	0.38	0.24	0.42		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		+0.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page